Simon Schleich

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Summary

I am an Associated Researcher at the Institute for Astrophysics at the University of Vienna. I research the atmospheres of exoplanets by performing data reduction on spectroscopic observations taken with the *James Webb* Space Telescope (JWST) and by interpreting these datasets using Bayesian parameter inference.

Education

PhD (Dr. rer. Nat., Astronomy) 10/2022 – 07/2026
University of Vienna Vienna, Austria
Thesis: Challenging Atmospheric Retrievals (Full text)
Supervisor: S. Boro Saikia, M. Güdel, A. Voigt

MSc (Master of Science, Astronomy) 10/2019 – 01/2022
University of Vienna (*Graduated with distinction*) Vienna, Austria
Thesis: Stellar wind simulations in the age of PSP and SolO (Full text)
Supervisor: S. Boro Saikia

BSc (Bachelor of Science, Astronomy) 10/2016 – 09/2019
University of Vienna Vienna, Austria
Thesis: Evolution and escape of atmospheres of planets orbiting M dwarfs (Full text)
Supervisor: K. Kislyakova

Research experience

PhD project 10/2022 – 12/2025
Thesis title: Challenging atmospheric retrievals
Characterising exoplanet atmospheres from JWST observations. End-to-end data reduction and application of atmospheric retrieval techniques, with a focus on NIR transmission spectroscopy. Results have been published in my thesis, as well as in Schleich et al. (2024), Schleich et al. (2025), and Schleich et al. (submitted)

MSc project 10/2020 – 12/2021
Thesis title: Stellar wind simulations in the age of PSP and SolO
Investigated the properties of the solar wind using both numerical simulations and observation. The project included (1) implementation of a WTD heating mechanism into the 3D MHD code NIRVANA (code development), and (2) data analysis of PSP and SolO measurements. Results have been published in my thesis and Schleich et al. (2023).

Select publications

Five peer-reviewed publications (three published as first author), three publications in the submission process.

– **Schleich, S.**, Boro Saikia, S., Changeat, Q., Güdel, M., Voigt, A. and Waldmann, I. (2025). "*Knobs and dials of retrieving JWST transmission spectra. II. Impacts of pipeline-level differences on retrieval posteriors*". A&A, 704, A223

– **Schleich, S.**, Boro Saikia, S., Changeat, Q., Güdel, M., Voigt, A. and Waldmann, I. (2024). "*Knobs and dials of retrieving JWST transmission spectra: I. The importance of p–T profile complexity*". A&A, 690, A336.

– **Schleich, S.**, Boro Saikia, S., Ziegler, U., Güdel, M. and Bartel, M. (2023). "*NIRwave: A wave-turbulence-driven solar wind model constrained by PSP observations*". A&A, 672, A64.

Select conference presentations

Below is a list of select national and international conference presentations

– *Retrieving p-T structure in panchromatic transmission spectra*
Talk: **Ariel Consortium Meeting**, Vienna, Austria, September 2026 (Slides)

– *Impacts of pipeline-level differences on atmospheric retrievals*
Poster: **Science with Hubble and JWST VIII**, Vienna, Austria, April 2026 (Poster)

– *Retrieving pressure-temperature profiles from exoplanet transmission spectra*
Talk: **Ariel Consortium Meeting**, Lisbon, Portugal, October 2024 (Slides)

Teaching

- Co-supervision of 2 BSc theses
- Co-supervision of 1 ERASMUS+ student
- 2025S: Dynamics of Gravitating Systems (BSc course, exercise leader)
- 2024W: Stars and Planetary Systems (BSc course, exercise leader)
- 2024S: Radiative Processes & Quantum Mechanics (BSc course, exercise leader)
- 2023W: Stars and Planetary Systems (BSc course, exercise leader)

Other work

Public outreach

Speaker: Nights at the observatory (*Nachts auf der Sternwarte*) 2025
The sun, moon, and stars: Astronomy as a science and inspiration for music (*Sonne, Mond und Stenre: Astronomie als Wissenschaft und Inspiration in der Musik*, held in German). A recording is available on YouTube.

Speaker: Nights at the observatory (*Nachts auf der Sternwarte*) 2025
New horizons on extrasolar planets (*Neue Horizonte auf extrasolaren Planeten*, held in German). A recording is available on YouTube.

Speaker: Long Night of Research (*Lange Nacht der Forschung*) 2024
Observing exoplanets with space telescopes (*Beobachtungen von Exoplaneten mit Weltraumteleskopen*, held in German).

Conference and event organisation

LOC member: Science with the Hubble and JWST VIII 2026
ESA-sponsored conference, in collaboration with STScI, highlighting the scientific impact of the Hubble and James Webb Space Telescopes. Held in Vienna.

Co-organiser: Embracing Diversity in STEM 2025
Inter-faculty event organised together with the Faculty for Chemistry, Mathematics, and Physics, with approximately 200 participants (in-person).

SOC member: Wienerwald Astronomy Symposium 2024
Joint symposium between the Department of Astrophysics at the University of Vienna and the Astronomy Research Groups at the Institute for Science and Technology Austria (ISTA).

Co-organiser: Big Picture Talks and Events 2023
Graveyard of Space Technology, organised within the VISESS Big Picture Talk framework, with approximately 100 participants (online and in-person)

Memberships

- Ariel consortium
  Mission science consortium
  (since 2023)
- EAS
  European Astronomical Society
  (since 2023)
- ÖGAA
  Austrian Society for Astronomy and Astrophysics
  (since 2023)
- VISESS PhD Board
  Vienna International School for Earth and Space Science
  (2023 – 2025)

Grants

VISESS Completion Grant – VISESS 2025
Research support awarded for the completion of the final phase of a PhD project.

Early-stage Researchers Programme – FGGA 2025 / 2024 / 2023
Travel grant awarded for scientific contributions to the annual EAS meeting (2025, poster contribution), Exoplanets 5 (2024, poster contribution), and PLANET ESLAB (2023, poster contribution).

International Communication – Austrian Research Federation (ÖFG) 2024
Travel grant awarded for a research visit to the ESA office at STScI (Baltimore, USA).

Short-term research stay abroad (KWA) – University of Vienna 2024
Travel grant awarded for a research visit to the ESA office at STScI (Baltimore, USA).

Performance scholarship (*Leistungsstipendium*) – University of Vienna 2021 / 2020
Granted for excellence in academic achievement.

Skills

Languages: English (fluent), German (native)

Programming: Python (experienced), C (familiar), HPC (Slurm Workload Manager)

Tools: *Eureka!* (JWST data reduction), TauREx (atmospheric retrieval), Git, LaTeX

List of publications

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[1] **Schleich, S.**, Boro Saikia, S., Changeat, Q., Güdel, M., and Voigt, A. (submitted). "*Knobs and dials of retrieving JWST transmission spectra. III. Panchromatic p–T preferences*". Under review in A&A.

[2] Changeat, Q., Lagage, P-O., Tinetti, G., Edwards, B., Charnay, B., Cowan, N. B., Danielski, C., Ducrot, E., Dyrek, A., Ikoma, M., Lueftinger, T., Micela, G., Morello, G., Panic, O., Pascale, E., Robert, S., **Schleich, S.**, Venot, O., Barstow, J. K., Bocchieri, A., Cho, J. Y-K., Cloutier, R., Coustenis, A., Dang, L., Fujii, Y., Ito, Y., Lavvas, P., Miguel, Y., Mugnai, L. V., Yip, K. H., and Zak, J. (submitted). "*On the synergetic use of Ariel and JWST for exoplanet atmospheric science*". RASTI (accepted for publication).

[3] Boro Saikia, S., Tennyson, J., Ji, S., Robeling, N.-M., Stankovic, I., Van Looveren, G., **Schleich, S.**, Johnstone, C., Kislyakova, K., and Güdel, M. (2026). "*The physical and chemical structure of Titan's upper atmosphere. Energy balance, photochemistry, and Jeans escape*". MNRAS (accepted for publication).

[4] Robeling, N.-M., Boro Saikia, S., Van Looveren, G., Stankovic, I., **Schleich, S.**, Johnstone, C., Güdel, M., and Kislyakova, K. (2026). "*Self-consistent 1D modelling of Jupiter's upper atmosphere as an exoplanet analogue*". A&A, 713, A260

[5] **Schleich, S.**, Boro Saikia, S., Changeat, Q., Güdel, M., Voigt, A. and Waldmann, I. (2025). "*Knobs and dials of retrieving JWST transmission spectra. II. Impacts of pipeline-level differences on retrieval posteriors*". A&A, 704, A223

[6] Van Looveren, G., Boro Saikia, S., Herbort, O., **Schleich, S.**, Güdel, M., Johnstone, C. and Kislyakova, K. (2025). "*Habitable Zone and Atmosphere Retention Distance (HaZARD): Stellar-evolution-dependent loss models of secondary atmospheres*". A&A, 694, A310.

[7] **Schleich, S.**, Boro Saikia, S., Changeat, Q., Güdel, M., Voigt, A. and Waldmann, I. (2024). "*Knobs and dials of retrieving JWST transmission spectra: I. The importance of p–T profile complexity*". A&A, 690, A336.

[8] **Schleich, S.**, Boro Saikia, S., Ziegler, U., Güdel, M. and Bartel, M. (2023). "*NIRwave: A wave-turbulence-driven solar wind model constrained by PSP observations*". A&A, 672, A64.

List of conference presentations

[1] *Retrieving p-T structure in panchromatic transmission spectra*
Talk: **Ariel Consortium Meeting**, Vienna, Austria, September 2026 (Slides)

[2] *Impacts of pipeline-level differences on atmospheric retrievals*
Poster: **Science with Hubble and JWST VIII**, Vienna, Austria, April 2026 (Poster)

[3] *Impacts of pipeline-level differences on atmospheric retrievals*
Poster: **Ariel Open Conference**, Harwell, UK, March 2026 (Poster)

[4] *Sensitivity in atmospheric retrievals of JWST hot Jupiter transmission spectra*
Poster: **EAS Annual Meeting**, Cork, Ireland, June 2025 (Poster (Online))

[5] *Retrieving pressure-temperature profiles from exoplanet transmission spectra*
Talk: **Ariel Consortium Meeting**, Lisbon, Portugal, October 2024 (Slides)

[6] *Influences of data processing techniques on the interpretation of atmospheric spectra from JWST*
Poster: **Exoplanets 5**, Leiden, The Netherlands, June 2024 (Poster)

[7] *Influences of data processing techniques on the interpretation of atmospheric spectra from JWST*
Talk: **Austrian Early Career Conference**, Salzburg, Austria, March 2024 (Slides)

[8] *NIRwave: A wave-turbulence-driven solar wind model constrained by PSP observations*
Poster: **Planet ESLAB 2023**, ESTEC, The Netherlands, March 2023 (Poster)